

Corrigé — Examen d'entraînement LEPL1110 — Éléments Finis

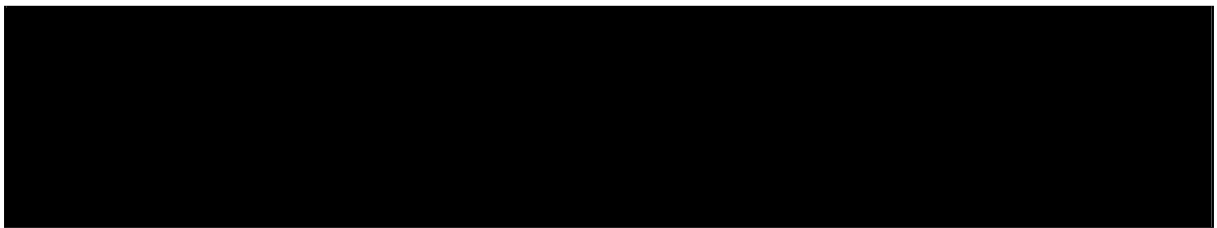
Claude code

Mai 2026

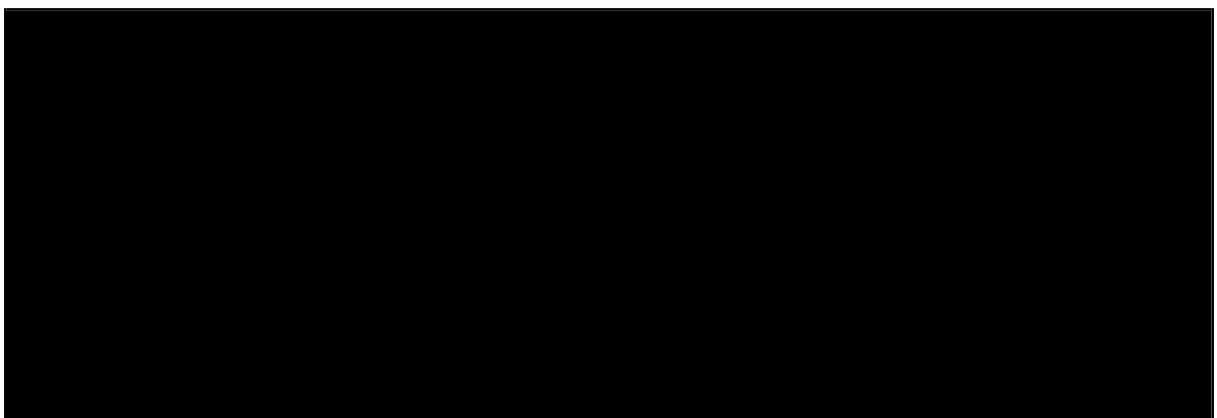
Exercice 1 — Formulation variationnelle, conditions mixtes et analyse d'erreur

1.1 — Espace fonctionnel et formulation faible

(a)

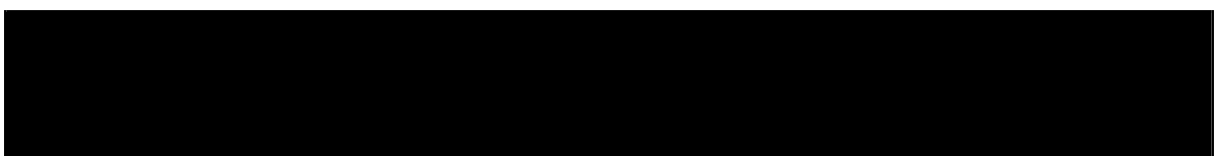


(b)



1.2 — Application du théorème de Lax-Milgram

(a)



(b)

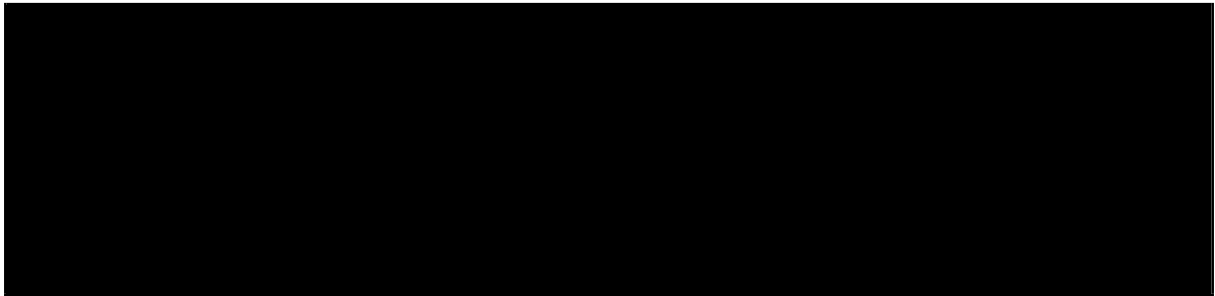


(c)



1.3 — Discrétisation et estimation d'erreur

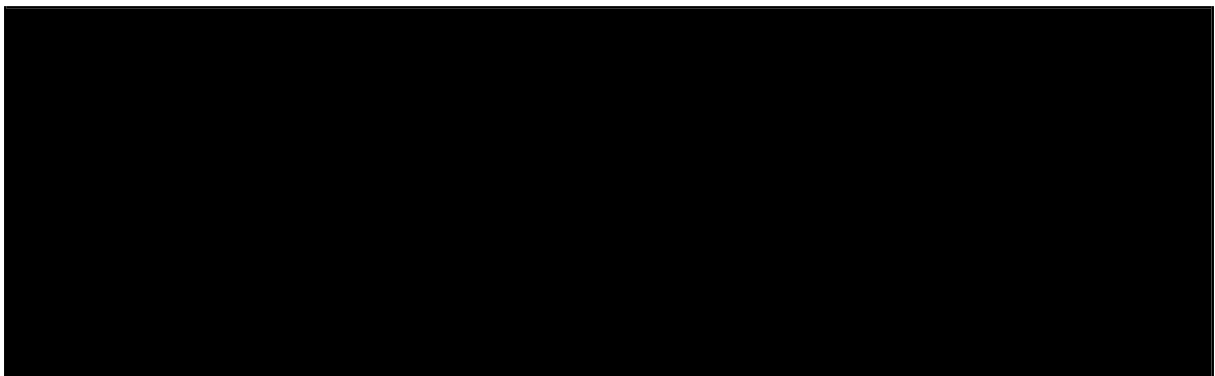
(a)



(b)



(c)



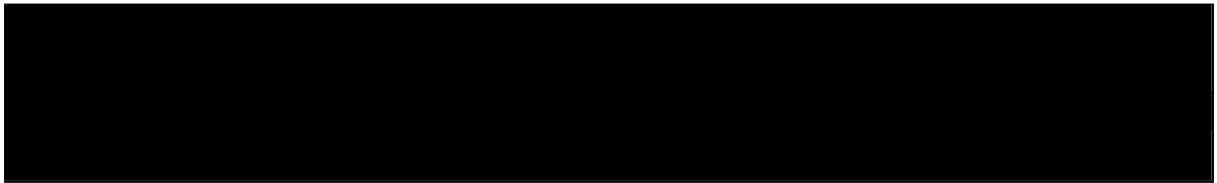
Exercice 2 — Éléments isoparamétriques Q1

2.1 — Propriétés des fonctions de forme

(a)



(b)



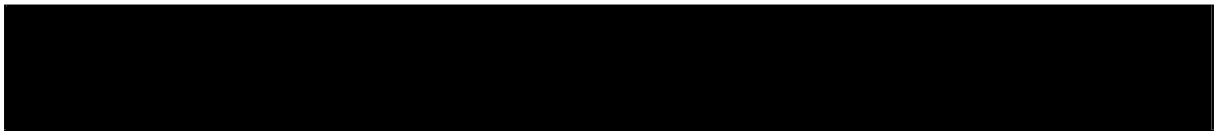
$= 1.$

2.2 — Application isoparamétrique

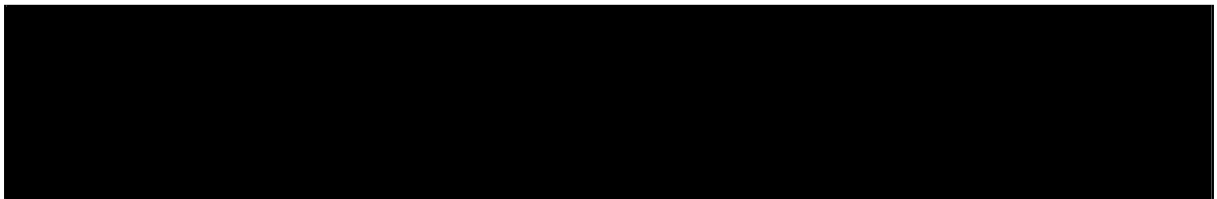
(a)



(b)

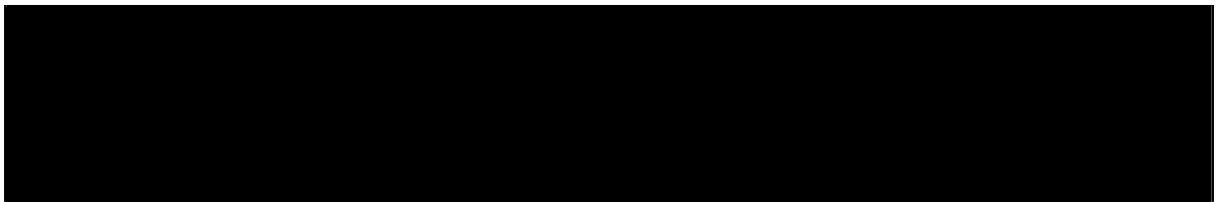


(c)



2.3 — Matrice de rigidité

(a)



(b)



2.4 — Matrice de masse et masse condensée

(a)



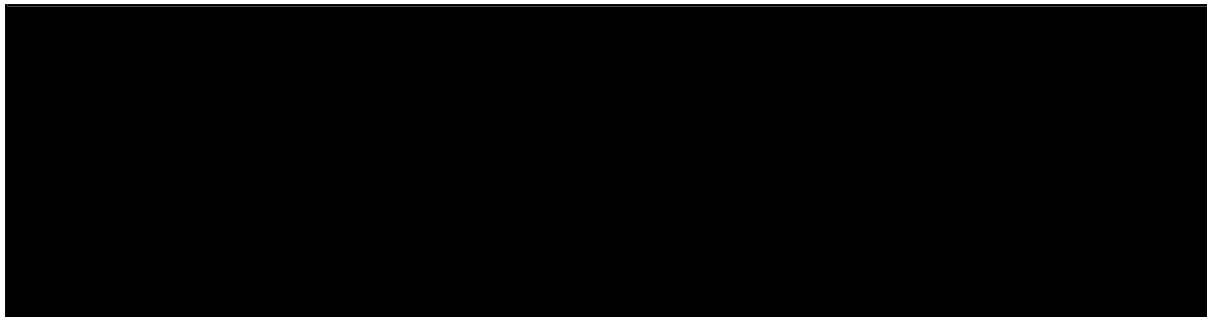
(b)



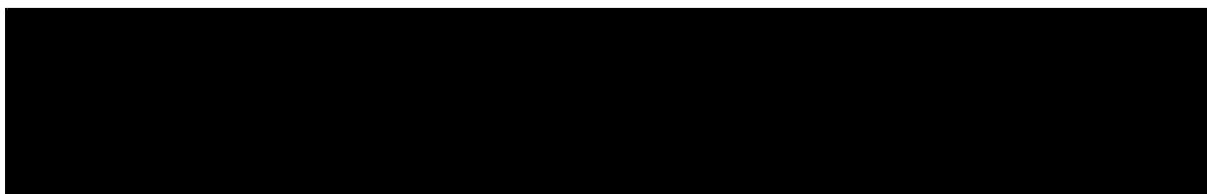
Exercice 3 — Diffusion instationnaire

3.1 — Semi-discrétisation spatiale

(a)

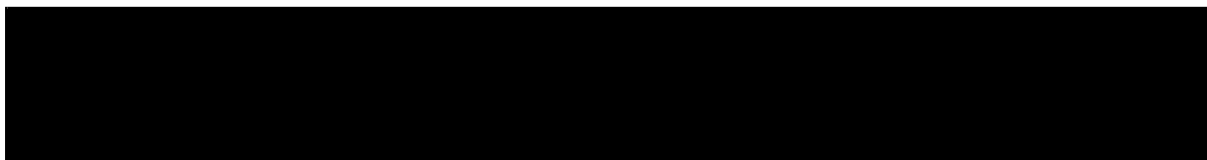


(b)

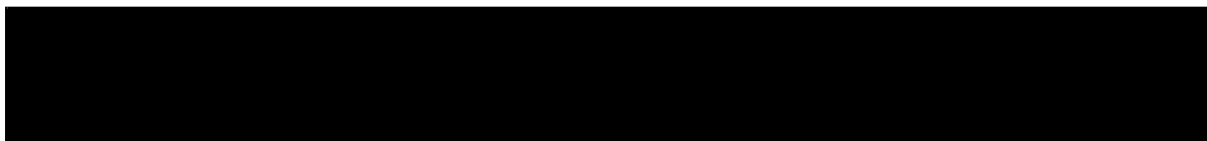


3.2 — Schéma θ

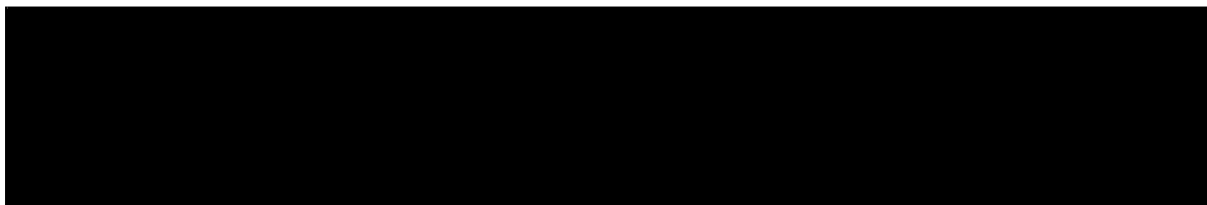
(a)



(b)



(c)



3.3 — Stabilité

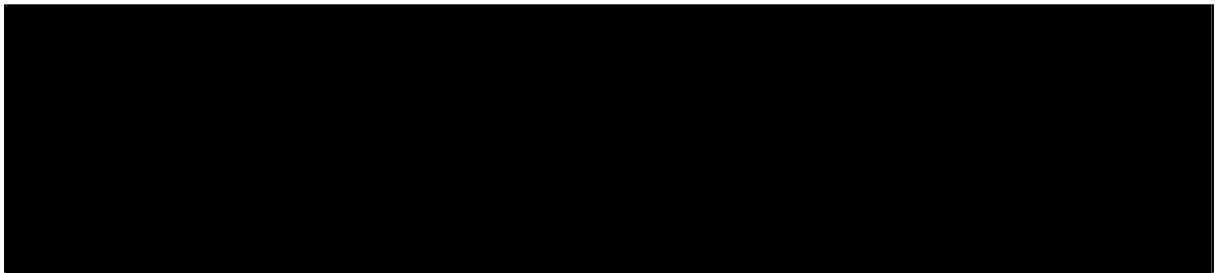
(a)



(b)

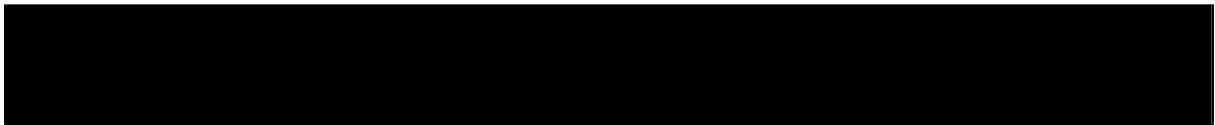


(c)



3.4 — Convergence globale espace-temps

(a)



(b)

